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Topic of Study

Climate and Economic Determinants of Child Malnutrition in East Africa

Research Question

How are rising droughts and changing weather patterns shaping child malnutrition trends in East Africa between 2010 and 2024

Full Citation and Link to Scientific Study

Thiede, Brian C., and Johann Strube. "Climate Variability and Child Nutrition: Findings From sub-Saharan Africa." *Global Environmental Change*, vol. 65, Nov. 2020, p. 102192.

<https://doi.org/10.1016/j.gloenvcha.2020.102192>.

Description of the Project

Introduction to the Research Area

Malnutrition among children under five remains a serious public health issue in East Africa. Many households depend on agriculture that relies heavily on rainfall, leaving them vulnerable to droughts and unpredictable weather. When rainfall drops or temperatures rise, food production declines, which often leads to nutritional deficiencies among children. This project focuses on understanding how rainfall and temperature trends relate to child malnutrition outcomes in Kenya, Ethiopia, and Tanzania between 2010 and 2024.

Introduction to the Scientific Dataset

This study builds on the work of Thiede et al. (2020), which examined climate and nutrition relationships across sub-Saharan Africa using the Demographic and Health Surveys (DHS) and climate data from the University of East Anglia's Climatic Research Unit (CRU). Our project narrows the focus to East Africa and includes more recent data.

The CRU TS v4.09 dataset contains monthly precipitation and temperature data from 1901 to 2024. We will extract and use only the data from 2010 to 2024 for Kenya, Ethiopia, and Tanzania to match the period of interest. The DHS Children's (KR) files will provide data on child malnutrition indicators such as height-for-age, weight-for-height, and weight-for-age, which measure stunting, wasting, and underweight. The DHS data will also include socioeconomic details such as the mother's education, household wealth, and place of residence (urban or rural).

Planned Visualizations and Analysis

Once DHS approval is finalized, both datasets will be merged by country and survey year. Before merging, we will perform data cleaning to remove missing or duplicate values and drop unnecessary columns that do not contribute to the analysis. We will also standardize date formats and variable names so that both datasets align properly. The analysis will be conducted in Python using Jupyter Notebook.

Planned visualizations include line graphs showing rainfall and temperature changes, scatterplots comparing precipitation and malnutrition rates, and heatmaps showing how socioeconomic conditions interact with environmental factors. We also plan to include correlation matrices to identify strong relationships and may use regression analysis to observe long-term effects. Additional visuals, such as regional maps showing patterns of drought and malnutrition, may be added later. Final decisions about the analyses and visualizations will be made as the project develops and we assess data.

Expected Outcomes

We expect to find that years with reduced rainfall and higher temperatures correspond to higher rates of child malnutrition, especially in rural and low-income households. These results could strengthen the understanding of how environmental stress interacts with economic and social factors to affect child health. The findings may also inform public health and food security initiatives in the region.

Public Dataset Locations

Climatic Research Unit (CRU TS v4.09): <https://crudata.uea.ac.uk/cru/data/hrg/>

DHS Program (Children’s Datasets): <https://dhsprogram.com/data/available-datasets.cfm>

Data Size (Rows and Columns)

Dataset	Countries	Years	Columns	Rows
CRU TS v4.09 (Precipitation and Temperature)	Kenya, Ethiopia, Tanzania	2010–2024	3	540 rows (annual data from 3 countries × 15 years × 12 months)

				summarized by year)
DHS (Children's)	Kenya, Ethiopia, Tanzania	2010–2022	12	Around 61,300 total child records (Kenya: 18,500, Ethiopia: 26,000, Tanzania: 16,800)

References

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<https://doi.org/10.1016/j.gloenvcha.2020.102192>.

University of East Anglia. (2025). Climatic Research Unit Time Series (CRU TS v4.09).
<https://crudata.uea.ac.uk/cru/data/hrg/>

USAID Demographic and Health Surveys (DHS) Program. (2024). Available Datasets.
<https://dhsprogram.com/data/available-datasets.cfm>